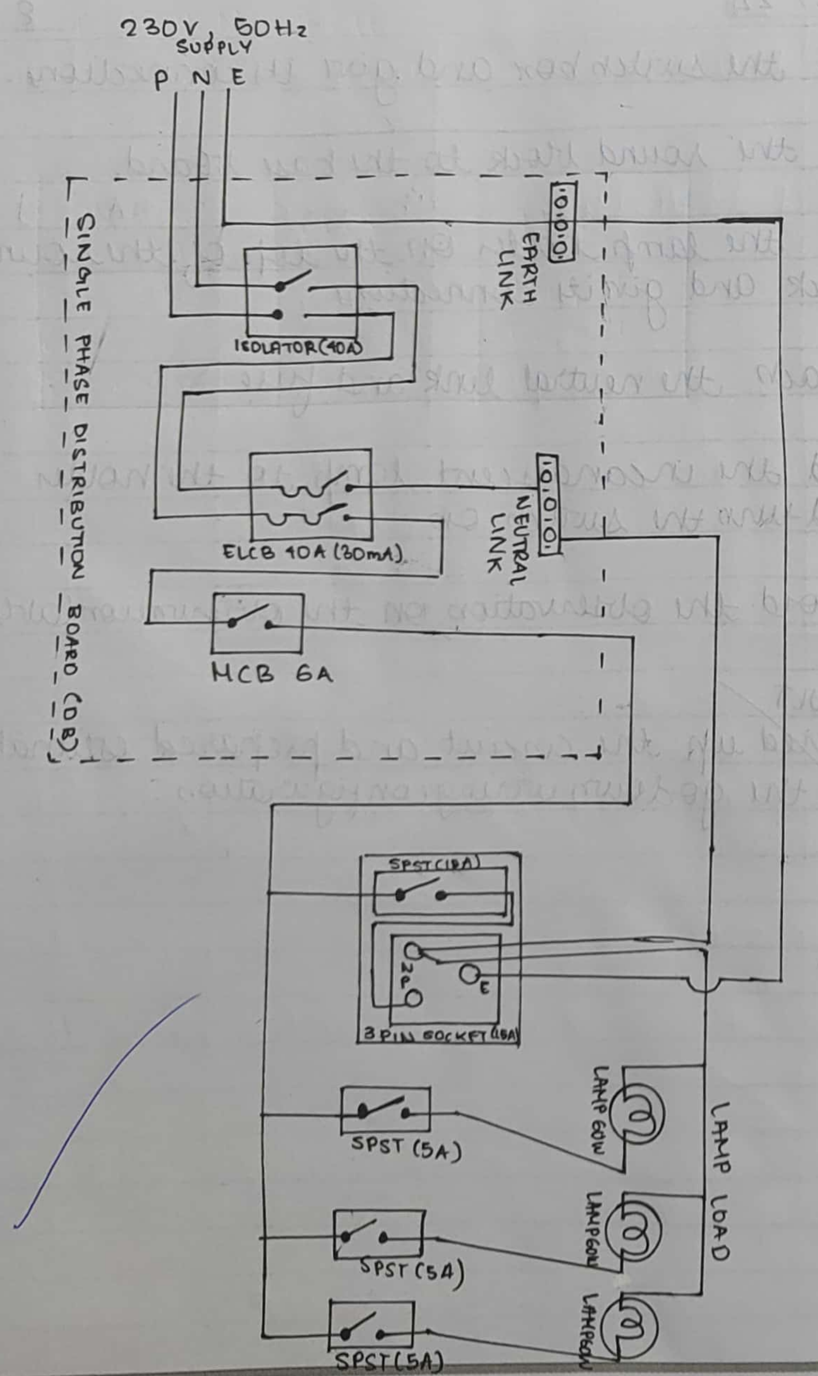

SINGLE PHASE DISTRIBUTION BOARD

AIM

Study the connection diagram and working of single phase distribution board.

THEORY

A distribution board is a panel or enclosure that houses the fuses, circuit breakers, and ground leakage protection units used to distribute electrical power to numerous individual circuits or consumer points. The board typically has a single incoming power source and includes a main circuit breaker and a residual current or earth leakage circuit breaker.

Distribution boards are common place in most industrial installation and commercial or residential buildings. Most consist of a panel or enclosure supplied with a single incoming electrical feed cable. The power is then split among several small circuit breakers or, in the case of older boards, fuses which in turn feed power to different consumption points or circuits. The core function of any distribution board is to allow individual circuits to draw power from correctly rated circuit breakers and for those circuit to be isolated without causing a disruption to the rest of the supply. Most

OBSERVATION TABLE

CONTACT BETWEEN	ELCB	MCB
Phase & Neutral	—	Tripped
Phase & earth	Tripped	Tripped
Earth & Neutral	Tripped	

importantly through, the distribution board offers protection to users and equipment from electrical shock or fire resulting from ground faults.

CIRCUIT BREAKERS

MCBs or Miniature circuit breakers are electromechanical devices which protect an electric circuit from an overcurrent. The over current, in an electrical circuit, may result from short circuit, overload or faulty design. An MCB is a better alternative to a fuse since it does not require replacement once an overload is detected. Unlike fuse, an MCB can be easily reset and thus offers improved operational safety and greater convenience without incurring large operational cost. The principal of operation is quite simple. An MCB functions by interrupting the continuity of electrical flow through the circuit once a fault is detected. In simple terms MCB is a switch which automatically turns off when the current flowing through it passes the maximum allowable limit. Generally MCB are designed to protect against over current and over temperature faults. (overheating)

There are two contacts one is fixed and other is movable. When the current exceeds the predefined limit a solenoid forces the movable contact open (ie disconnect from the fixed contact) and the MCB turn off thereby stopping the current to flow in the circuit. In order to restart the flow of current the MCB is turned on manually. This mechanism is used to protect from the faults arising due to over current or over load. To protect against fault arising due to over heating or increase in temperature a bi-metallic strip is used. MCB's are generally designed to trip within 2.5 milli second. when an over current fault arises. In case of temperature rise or over heating it may take 2 seconds to 2 minutes for the MCB to trip.

EARTH LEAKAGE CIRCUIT BREAKERS

An earth leakage circuit breaker (ELCB) is a device used to directly detect currents leaking to earth from an installation and cut the power.

MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1)	Isolator	2 pole 1 ϕ	40 A	1
2)	ELCB	2 pole 1 ϕ	40A (30mA)	1
3)	MCB	–	6A/10A	1
4)	earth link	Porcalin base with copper segment	16A	1
5)	Neutral link	Porcalin base with copper segment	16A	1
6)	switch	SPST	5A	3
7)	lamp	Incandescent	60W, 250V	3
8)	switch	SPST	15A	1
9)	socket	3 Pin	15A	1
10)	DB	6 way	240/450V	1
11)	steady batton holder		5A	3

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12)	Round block PVC	3 inch		3
13)	Switch box	6x4"		1
		3x2"		3

TOOLS REQUIRED

Screw driver, line tester, combination plaier ball-peen hammer, poker, nose plier, wire stripper

PROCEDURE

connect

i) Phase and neutral in contact

ii) Phase and earth in contact

iii) Neutral and earth in contact and check the observation in ELCB & MCB.

RESULT

studied and verified the connection diagram and working of single phase distribution board.

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